DAY 12 | POST-READ

GenAI, LLMs and Tokens: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
LLM loop	Tokenise, embed, transform with attention, predict the next token, repeat.
Attention	softmax(QK^T / sqrt(d)) V: each token weighs the others.
Architectures	Encoder-only (understand), decoder-only (generate), encoder-decoder (transform).
Multimodal and computer use	Models read images and audio; some can operate a sandboxed screen.
Reasoning models (NEW v4)	Think before answering; reasoning effort trades cost for quality.
Open-weight and SLMs (NEW v4)	Run on your own hardware; cheap and private, narrower skills.
Tokens (NEW v4)	Input, output, reasoning and cached tokens are priced differently.
First API call	Key from Secrets or .env, model from LLM_MODEL, log usage for every call.

Part B: Going beyond today

1. Mixture of experts (MoE)

A router sends each token to a few specialist sub-networks. The model has huge capacity but uses only part of it per token, so cost per token stays lower.

2. Temperature and sampling

Temperature reshapes the next-token probabilities. Low values give focused, repeatable output. Reasoning models may fix or ignore temperature; check the API documentation for the model you use.

3. Speed tiers and batching

Providers offer faster tiers at a higher price and batch processing at a lower price for jobs that can wait. Pick by how quickly the user needs the answer.

4. Prompt caching

When many requests share the same opening (system prompt, tool list, policy text), the provider can cache it and charge less for those tokens. Keep stable text at the start of the prompt. More on Day 13 and Day 25.

5. Counting before you call

Count tokens with tiktoken before sending a large document. If it will not fit, or will cost too much, trim or retrieve only the relevant part. That is the idea behind RAG on Day 18.

Part C: Check yourself

- Why does an LLM sometimes give different answers to the same question?
- In your own words, what does attention do?
- Why can a reasoning model cost more even when its visible answer is short?
- When would you choose a small open-weight model over an API LLM?
- Why should the API key never be in front-end code?

Part D: Practice ideas for tonight

- Count tokens for the same sentence in English and Japanese. What do you notice?
- Estimate the monthly cost of summarising 5,000 tickets a day using the placeholder prices.
- Add a cost column to your SmartDesk token log.

Coming up on Day 13: prompt engineering, structured outputs and GenAI assessment 1.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today